

AI-driven Risk and Scope Management System

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Introduction

The growth of technology projects and the increasing complexity of project environments made risk and scope management more critical than ever. Many projects fail because of unclear scope, unmanaged risks and poor planning. This project aims to develop an AI Project Risk & Scope Management System that uses artificial intelligence techniques to help organizations monitor risks and manage project scope

and support better decision making during the project lifecycle.

The proposed system aims to provide a centralized dashboard that analyzes project data, detects scope creep, identifies schedule delays, and monitors resource allocation. Additionally, the system provides intelligent recommendations such as adjusting schedules, redistributing resources, and modifying project scope when necessary. This approach enhances project efficiency, reduces risks, and supports proactive project management.

The goal of the Project Charter is to define the project scope, stakeholders, objectives, assumptions, constraints and the overall direction of the project. It's an agreement between stakeholders and the project team to make sure everyone understands the project and their responsibilities.

The Work Breakdown Structure (WBS) is used to divide the project into smaller and manageable tasks which helps in better planning and resource allocation and tracking the progress of the project.

This project is designed to simulate a real world enterprise level solution supported by major stakeholders and intended for organizations that manage complex technology projects.

Project Background

Poor scope management, delays in timelines, and inefficient use of resources are some of the main reasons why projects fail. Project managers often face challenges such as tracking changes in scope, monitoring task completion, and identifying team members who are overloaded with work. These issues can lead to project delays, increased costs, and lower success rates.

As projects become more complex, there is a need for a simple solution that helps project managers track performance and monitor risks more effectively. The proposed AI Project Risk & Scope Management System aims to provide a centralized platform that supports project management, risk detection, and decision-making. The system will help users track tasks, monitor deadlines, and identify workload problems, which can improve overall project delivery.

Project Charter

1.Genral Information:

Project Title: **AI-driven risk and scope management system**

Project ID: **001**

sponsoring Organization: **SADAIA & Monsha'at**

Sponsor Representative: **Dr. Abdullah Al-Ghamdi**
Dr. Majid Al-Kassabi

Prepared by: **Joud Alrefaei**

Version: **1.0**

2. Project Stakeholders:

Position	Title/Name/ Organization	Phone	E-mail
Sponsor Representative	SDAIA - Dr. Abdullah Al-Ghamdi	+966508769415	president@sdaia.gov.sa
	Monsha'at - Dr. Majid Al-Kassabi	+9661125006400	president@monshaat.gov.sa
Project Manager	Joud Alrefaei	+966546774329	joud@gmail.com
Customer / User Representative(s)	Joud Project Managers, Risk Analysts	N/A	N/A
Companies Managing Technology Projects	Technology Implementation Partners	N/A	N/A
Risk Management Experts	Risk Analysts, Risk Officers	N/A	N/A
Backend Developers	Rafa'a Alharbi,	+966588451296	Rafaa@gmail.com
	Layan Alotaibi	+966564797360	Layan@gmail.com
Frontend Developers	Reema Alahmadi	+966569832560	<u>Reema@gmail.com</u>
Data Engineers	Wejdan Alsaedi,	+966580092946	Wejdan@gmail.com
	Rayan Alrashidi	+966567853360	Rayam@gmail.com
Data Scientists	Ghala alharbi	+9665606386392	Ghala@gmail.com
	Amal Ahmed	+9665546393056	Amal@gmail.com
Security Engineers	Hajer Alrashidi	+966580928378	Hajer@gmail.com
DevOps Engineers	Abdullah Alrashidi	+966564763920	Abdullah@gmail.com
UI/UX Designers	Jana Aljohani	+966588451296	jana@gmail.com
QA Engineers	Yara Alharbi	++966583579754	Yara@gmail.com

3.Executive Summary

Through an intelligent, user-friendly, and integrated AI-based system(a project change detection system), project managers and organizations that manage technology projects

will be allowed to control the project's scope, easily detect minor risk, and any issues before it can negatively effect the project by continuously analyzing project's data leading to improved project success rates , better decision that relying on data, reducing unexpected changes, real time insights

4.Project Purpose

4.1 Business Need / Problem

In modern projects, managing risks or even project scope is a major challenge. Many projects constantly change their scope to adapt to new requirements and satisfy stakeholders, leading to altered plans, loss of control over project progress, and ultimately, project failure.

Furthermore, many risks only become apparent after they have already had their negative impact and hindered project progress, making decision-making difficult. Because of these challenges, there is a strong need for systems that help monitor projects, predict risks, and detect any scope changes early by analyzing project data in real time and providing actionable insights.

The proposed system transforms project management from a reactive to a proactive process, enabling optimal decision-making. These problems are important because they directly and negatively affect projects, such as project failures, poor scope control, and delayed risk detection.

4.2 Business Objectives

Strategic Plan Element	Project Business Objectives
Improve project risk management	Achieving at least 70% prediction accuracy in identifying project risks.
Better project management	Implementing continuous project monitoring with real-time data tracking and reporting.
Control scope	Detecting range deviations with an accuracy of at least 60% in the early stages
Improve decision making	Improving decision-making speed and accuracy by at least 20% using AI-driven insights.
Increase stakeholder satisfaction	Increasing stakeholder satisfaction through improved project success rates and transparency
Efficient investment	Reducing cost overruns by at least 20%.
Optimal security	Enhancing security and technical governance compliance by at least 30%

5. Project Overview

5.1 Project Description

The AI Project Risk & Scope Management System is a web-based system designed to help project managers monitor project scope, schedule, and resource risks. The system allows users to create projects, assign tasks, define deadlines, and track project progress.

The system provides features for detecting delayed tasks, identifying overloaded team members, and monitoring scope changes. It also includes dashboards and reports to display project status and highlight potential risks. The system aims to improve project monitoring, enhance decision-making, and help ensure project completion within scope and schedule.

5.2 Scope

In Scope :

The project includes the development of an AI-based web system that enables project managers and organizations to monitor project performance and detect risks and scope changes.

The system will allow users to:

- Create and manage projects, tasks, timelines, and team members
- Monitor project scope and detect deviations from the approved plan
- Identify risks related to schedule delays and resource allocation
- Generate alerts and notifications for potential issues
- Provide AI-based predictions and recommendations for decision-making
- Display dashboards and reports showing project status and risk levels
- Store and manage project data securely using a database
- Support role-based access for different users (project managers, team members, administrators)

Out of Scope :

The project does not include:

- Full financial management or accounting features
- Replacement of human decision-making (AI provides support only)
- Integration with all external enterprise systems
- Development of a fully featured mobile application (focus is on web system)

5.3 Assumptions

1. It is assumed that project managers and team members will be willing to use the system effectively after receiving proper training and guidance.
2. It is assumed that sufficient project data will be available for the AI system to analyze and generate insights.
3. It is assumed that the AI model will provide reasonably reliable predictions based on available data.
4. It is assumed that the required technical infrastructure, including servers, databases, and network availability, will be stable to support system operations.
5. It is assumed that organizations will follow appropriate security practices to protect sensitive project data.

5.4 Constraints

1. The project development budget must not exceed 300,000 SAR to ensure cost efficiency.
2. The system must be developed and delivered within a maximum period of 12 months.
3. The system will initially target organizations in Saudi Arabia, especially project-based environments.
4. The application will be developed as a web-based system accessible.
5. The system has medium complexity due to the integration of AI, dashboards, and analytics features.
6. The system performance and accuracy are constrained by the quality, accuracy, and completeness of the input project data

6. Project Requirements

Functional Requirements

- The system should allow project managers to create and manage projects, tasks, timelines, team members, and resources.
- The system should monitor project scope changes and detect any task or feature that is outside the approved scope.
- The system should identify schedule risks by tracking delayed tasks, missed deadlines, and task dependencies.
- The system should detect resource risks such as overloaded team members or uneven workload distribution.
- The system should provide AI-based suggestions for solutions like adjusting the scope, rescheduling tasks, or reallocating resources.
- The system should send alerts and notifications when any abnormal situation happens in the project.
- The system should display dashboards and reports that show project progress, risk levels, and overall project status.
- The system should allow authorized users to update project data and review risk history.

Technical Requirement

- The system must be developed as a web-based application accessible through standard browsers.
- The system must include an AI or machine learning component to analyze project data and predict possible risks.
- The system must use a database to store project information, tasks, users, resources, risk logs, and recommendations.
- The system must support secure login and role-based access control for project managers, team members, and administrators.
- The system must provide acceptable performance when processing project updates and generating alerts.
- The system must be scalable to support multiple projects and users.
- The system must maintain data confidentiality, integrity, and backup capabilities.

Business Requirement

- The system must help organizations reduce project failure caused by scope creep, schedule delays, and poor resource allocation.
- The system must improve decision-making by providing early warnings and actionable recommendations.
- The system must enhance project visibility for managers and stakeholders.
- The system must support better use of time, budget, and human resources.

6. Project Deliverables

Project Management System (Web Application)

Scope Risk Detection Module

Schedule Risk Detection Module

Resource Risk Detection Module

Project Dashboard

Risk Alerts and Notification System

AI Recommendation Feature

Enables real-time decision making

Database Design

User Interface Design

System Testing and Validation

Project Documentation

7. Project Management Milestones and Deliverables

Milestone / Deliverable	Estimated Date	Responsible Individual
Project charter and requirements approval	Jan 15, 2027	Joud Alrefaei
Project Management & Data management plan completion	Jan 31, 2027	Joud Alrefaei
Data Readiness review completed	Feb 28, 2027	Rayan Alrashidi, Amal Ahmed
System design and architecture approval	Apr 10, 2027	Rafa'a Alharbi, Layan Alotaibi
AI Model performance validated	Jul 10, 2027	Ghala Alharbi, Amal Ahmed
system performance requirements met	Aug 31, 2027	Ghala Alharbi, Amal Ahmed
Deployment strategy and actionable results	Oct 10, 2027	Joud Alrefaei, Abdullah Alrashidi
User acceptance testing UAT completed and approved	Nov 15, 2027	Project Managers, Risk Analysts
Project closure	Dec 31, 2027	Joud Alrefaei

8 Costs / Budget

Purpose	Amount (SAR)	Source
Development team	180,000	SDAIA - Dr. Abdullah Al-Ghamdi Monsha'at - Dr. Majid Al-Qasabi
AI infrastructure and cloud services	60,000	
Data access and expert consultation	25,000	
Security implementation and licenses	20,000	
Contingency reserves	15,000	

9 Personnel & Other Resources

Resources	Description
Team Resources	· Joud Alrefaei · Rafa'a Alharbi · Layan Alotaibi · Reema Alahmadi · Wejdan Alsaedi · Rayan alrashidi · Ghala Alharbi · Amal Ahmed · Jana Aljohani · Yara Alharbi · Abdullah alrashidi · Hajer alrashidi
External Support	Risk management experts (Risk Analysts and officers), Technical partners
Infrastructure and computing	High performance computing resources for training deep learning models and big data handling.
Facilities	Collaborative Project environment (platforms for team collaboration and data handling)
Security implementation	Data Governance framework (implementation of encryption and access control to protect sensitive project data)
Software Tools	Professional tools for data engineering, AI modeling, and Business Intelligence to enable real-time proje monitoring.

10 Project Risks

Known Problems:

- **AI recommendations may require clear justification before using in project decisions**
Mitigation: provide explainable outputs and brief reasoning
- **Different organizations may use different project management practices**
Mitigation: support adaptable workflows and flexible settings
- **Maintaining consistent performance across different browsers and screen sizes may be challenging**
Mitigation: use responsive design and cross browser testing
- **Balancing system functionality with ease of use might be difficult**
Mitigation : prioritize essential features and simplify navigation

Anticipated Issues:

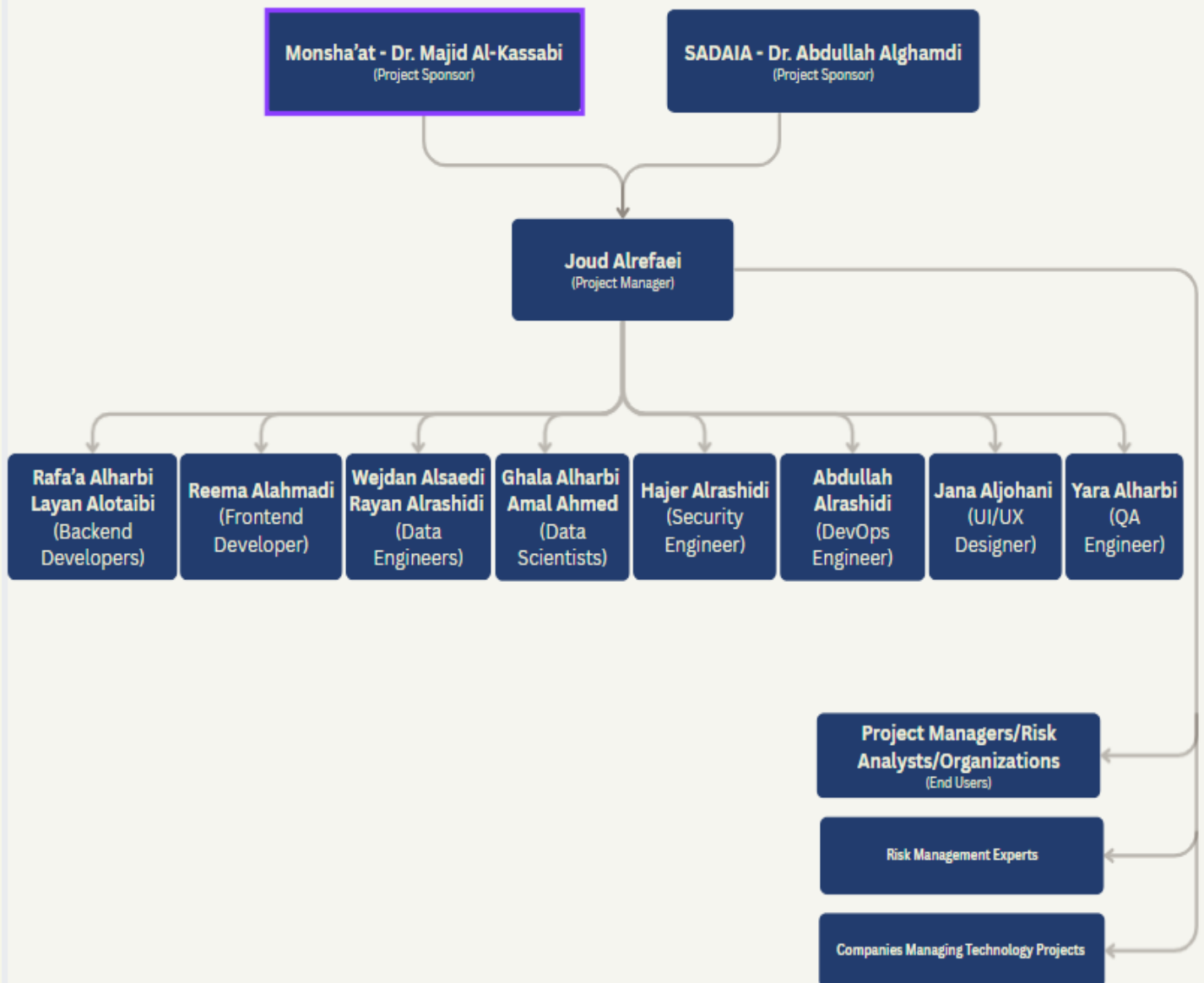
- **Realtime dashboards and analytics may require performance optimization**
Mitigation: optimize algorithms and test performance regularly
- **Different organizational technical environments may affect deployment and system configuration**
Mitigation : define technical requirements clearly and test representative deployment environments.
- **Users may expect system outputs and recommendations to align closely with actual project workflows**
Mitigation : Involve representative users in testing and refine outputs based on feedback
- **Security and controlled access will require ongoing attention**
Mitigation: apply authentication and encryption and access control
- **Technical bugs may appear during integration**
Mitigation: perform continuous testing and staged integration.

Potential Risks:

- **AI outputs may not always fit complex real project situations**
Mitigation: keep AI as decision support and require human review
- **System downtime may interrupt monitoring and reporting**
Mitigation: Use backup and recovery mechanisms
- **Security vulnerabilities may expose project data.**
Mitigation: apply secure coding and regular security testing.
- **Development delays may affect delivery within 12 months**
Mitigation: Monitoring milestones and resolve obstacles early.
- **Budget pressure may arise due to system complexity**
Mitigation: Control scope and prioritize essential deliverables

10 Project Organization

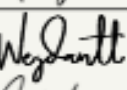
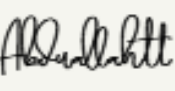
11.1 Project Organization chart



11.2 Roles & Responsibilities

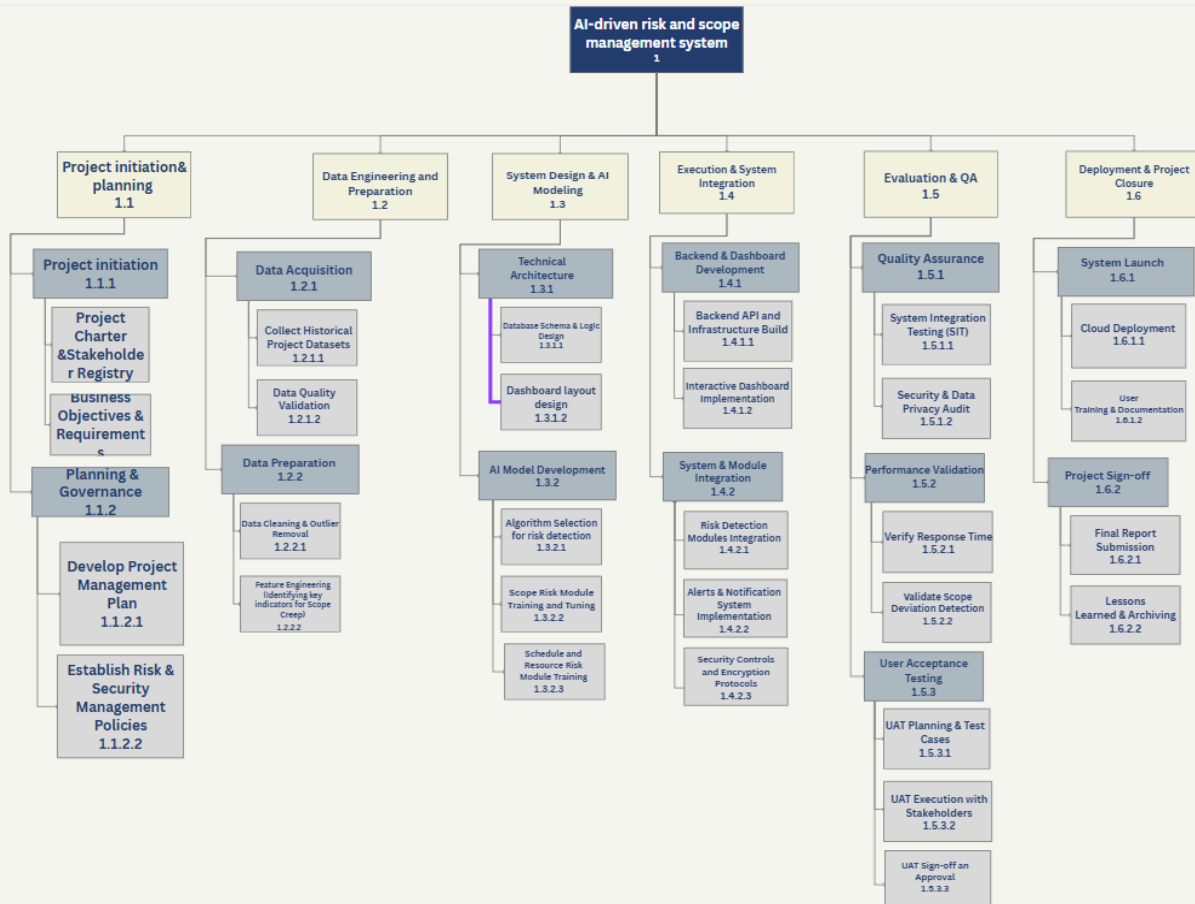
Stakeholder Title	Name	Roles & Responsibilities
Project Sponsor	Dr. Majid Al-Qasabi, Dr. Abdullah Al-Ghamdi	-sponsoring the project. -Define project goals and vision -Granting approvals for key decisions and deliverables - Evaluating project success
Project Manager	Joud Alrefaei	-Monitor scope changes using AI outputs - Check risk alerts generated by AI models -Coordinate between AI team and risk experts -Communicate with stakeholders and clients -Ensure project delivery
Backend Developers	Rafa'a Alharbi, Layan Alotaibi	-Build and maintain server-side logic and system functionality. -Build the core system logic -Integrate AI models into the system
Frontend Developers	Reema Alahmadi	-Develop user interfaces and features. -Connect frontend with backend services
Data Engineers	Wejdan Alsaedi, Rayan alrashidi	-Manage data infrastructure and pipelines. -Collect the data -Ensure data quality -Prepare data for ML models
Data Scientists	Ghala Alharbi, Amal Ahmed	-Analyze data , generate and train AI models for scope creep detection -Evaluate model performance and optimize machine learning algorithms -Analyze risk patterns and trends -Provide insights for decision-making
UI/UX Designers	Jana Aljohani	- Design dashboards for risk alerts and scope changes -Present AI outputs in an user-friendly way -Improve usability
QA Engineers	Yara Alharbi	- Test the accuracy of AI predictions - Ensure reliability and quality of risk alerts
DevOps Engineers	Abdullah alrashidi	- Deploy AI models - Monitor model performance
Security Engineers	Hajer alrashidi	- Secure sensitive data used in AI training - Ensure compliance with data privacy regulations-
End Users	Project Managers, Risk Analysts	-Individuals who will use the system. -Provide feedback on usability -Report issues and improvements
Risk Management Experts	Risk Analysts, Risk Officers	-Identifying and classifying risks -Providing the team with historical risk data - - Ensuring the accuracy of AI system results - - Developing risk management policies
Companies Managing Technology Projects	Technology Implementation Partners	-External organizations involved in project execution. -Ensure alignment with industry standards -Provide technical expertise. -Ensure alignment with technical standards

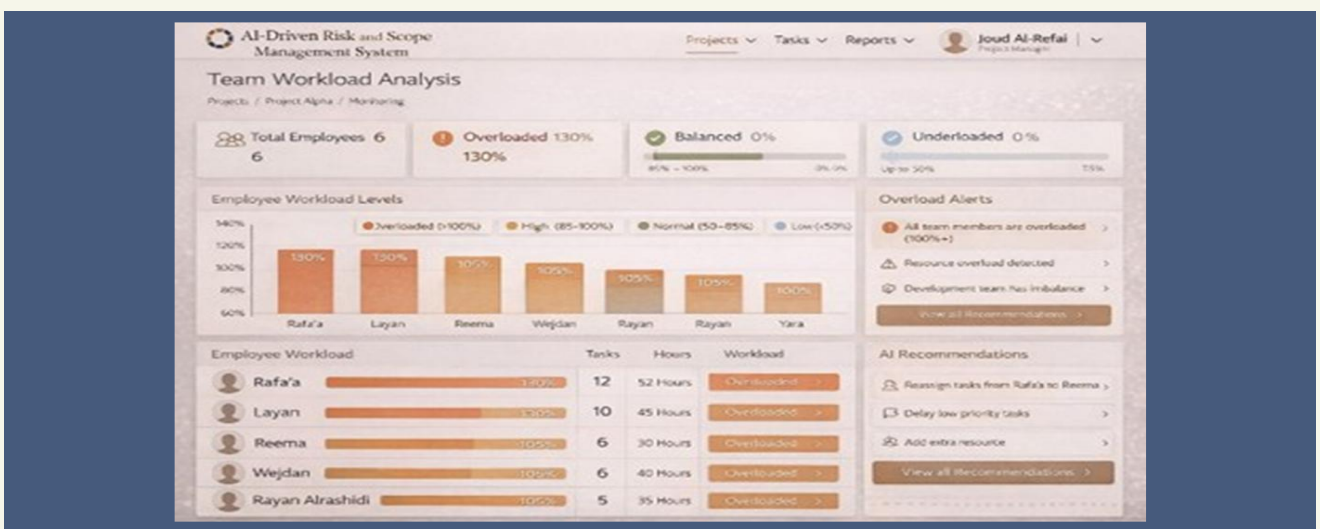
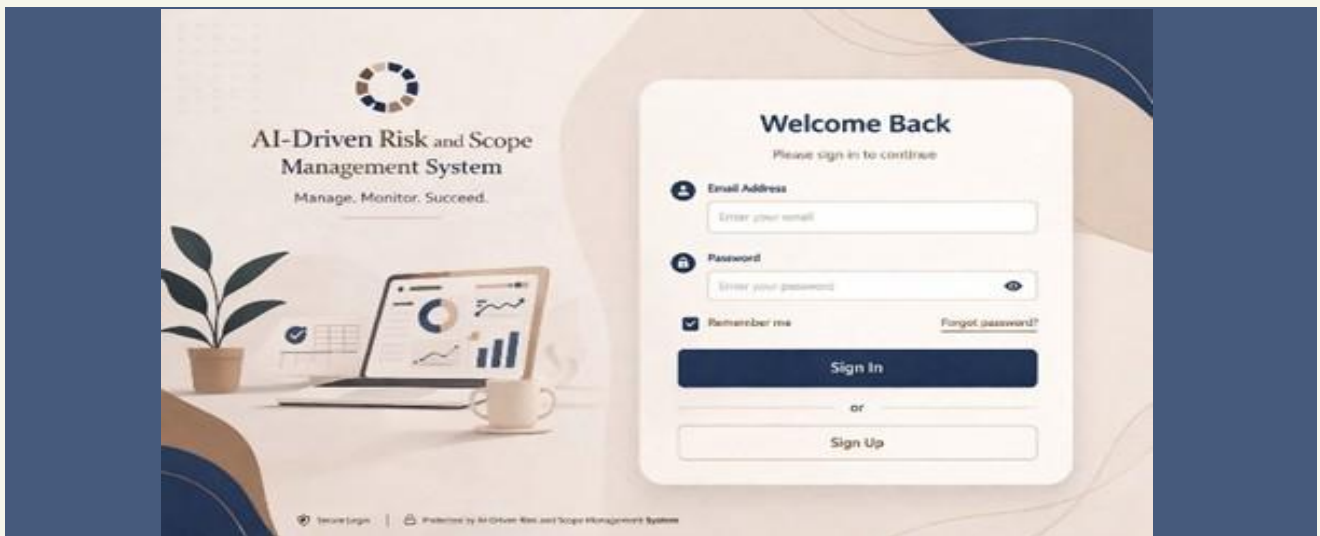
12.Approval Signatures

Position Title	Signature/Printed Name/Title	Date
Project Sponsor Representative	Dr. Majid Al-Qasabi 	5-4-2026
	Dr. Abdullah Al-Ghamdi 	
Project Manager	Joud Alrefaei 	1-4-2026
Backend Developers	Rafa'a Alharbi 	3-4-2026
	Layan Alotaibi 	
Frontend Developers	Reema Alahmadi 	3-4-2026
Data Engineers	Wejdan Alsaedi 	3-4-2026
	Rayan alrashidi 	
Data Scientists	Ghala Alharbi 	3-4-2026
	Amal Ahmed 	
UI/UX Designers	Jana Aljohani 	3-4-2026
QA Engineers	Yara Alharbi 	3-4-2026
DevOps Engineers	Abdullah alrashidi 	3-4-2026
Security Engineers	Hajer alrashidi 	3-4-2026
End User Representative	Khalid Alharbi 	5-4-2026

Part 2 : WBS

Hierarchical WBS:





Conclusion

12. Approval Signatures

The AI-Powered Risk and Scope Management System aims to improve project management efficiency by providing an intelligent platform that enables project managers to monitor project progress and identify potential risks at an early stage. The system focuses on analyzing key project risks, including scope changes, schedule delays, and resource overload, using artificial intelligence techniques to generate accurate predictions and actionable recommendations that support effective decision-making.

The proposed system enhances project control by reducing the likelihood of failure caused by unplanned scope changes, delayed activities, or improper resource allocation. It also offers interactive dashboards and real-time alerts that allow stakeholders to continuously track project performance and take timely corrective actions when needed. These capabilities improve visibility, communication, and overall project coordination.

Ultimately, the AI-Powered Risk and Scope Management System contributes to improving the quality of project management, increasing project success rates, and enabling data-driven decision-making. By providing early warnings and intelligent insights, the system helps ensure that projects are delivered within the defined scope, planned schedule, and available resources with greater efficiency and reliability.